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Impact of anthropogenic activities on the physico-chemical properties of Latratu Dam, Jharkhand, India

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Abstract- The present study evaluates the impact of anthropogenic activities on the physico-chemical properties of Latratu Dam, Jharkhand, India, during 2024 with emphasis on seasonal variability and ecological implications. Key parameters including temperature, pH, dissolved oxygen (DO), free carbon dioxide (CO_2), alkalinity, chloride, sulphate, and phosphate were analyzed using standard methods. Seasonal analysis revealed significant variation ($p < 0.05$), particularly during the monsoon period. Phosphate concentrations (0.02-0.25 mg L^{-1}) and CO_2 levels (4-18 mg L^{-1}) showed a consistent increase during monsoon, indicating nutrient influx from agricultural runoff and catchment disturbances. Dissolved oxygen exhibited a negative correlation with temperature and CO_2, reflecting increased biological oxygen demand associated with organic inputs. Chloride and sulphate levels, although within permissible limits, showed moderate enrichment, suggesting diffuse anthropogenic sources. The reservoir exhibits mesotrophic characteristics with early signs of nutrient-driven ecological alteration. These findings demonstrate that even low-intensity human activities can significantly influence water quality through seasonal pathways. Continuous monitoring and watershed-level management are recommended to mitigate further deterioration

Keywords: Latratu Dam, anthropogenic impact, water quality, physico-chemical parameters, nutrient enrichment

INTRODUCTION

Freshwater ecosystems are dynamic and highly responsive to both natural variability and anthropogenic disturbances. Physico-chemical parameters such as temperature, pH, dissolved oxygen, and nutrient concentrations regulate ecosystem metabolism, productivity, and trophic status. Alterations in these parameters, particularly due to human activities, can significantly disrupt ecological balance.

Anthropogenic inputs such as agricultural runoff, domestic discharge, and land-use changes contribute to nutrient enrichment and organic loading in aquatic systems, often leading to eutrophication.[1,2] In tropical reservoirs, these effects are frequently amplified during monsoon due to increased surface runoff and catchment transport processes.[3,4]

Latratu Dam is located in a relatively less industrialized region of Jharkhand, recent expansion of agriculture, fisheries, and local habitation suggests increasing anthropogenic pressure. Limited studies have examined how these activities influence seasonal water quality dynamics in such semi-forested reservoirs of eastern India.

The present study aims to evaluate the seasonal variation in key physico-chemical parameters of the

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reservoir, examine the interrelationships among these variables, and assess the extent to which anthropogenic activities influence overall water quality.

MATERIALS & METHODS

Study Area

Latratu Dam is situated in the Lapung region of Ranchi, Jharkhand, India. The reservoir is surrounded by a forest-dominated catchment but also experiences increasing human interference in the form of agricultural activities, domestic usage, and localized resource exploitation. The reservoir serves multiple purposes including irrigation, fisheries, and domestic use, making it socio-economically important. Seasonal rainfall significantly influences hydrological and chemical dynamics, particularly through runoff processes.

Sampling

Water samples were collected seasonally (winter, summer, monsoon) from multiple sampling locations representing littoral and open-water zones during 2024. Samples were collected in triplicates to ensure analytical reliability.

Analytical Methods

All physico-chemical parameters were analyzed following standard methods[5,6]:

- Temperature: Thermometric method
- pH: Digital pH meter
- Dissolved Oxygen (DO): Winkler's method
- Free CO_2: Titrimetric method
- Alkalinity: Acid titration
- Chloride: Argentometric method
- Sulphate: Colorimetric method
- Phosphate: Spectrophotometric method

Statistical Analysis

Data were analyzed using descriptive statistics and inferential tools. One-way ANOVA was applied to assess seasonal variation, while Pearson correlation analysis was used to examine relationships among parameters such as DO, temperature, CO_2, and nutrients.

RESULTS & DISCUSSION

Temperature and Dissolved Oxygen Dynamics

Temperature varied from 18°C in winter to 32°C in summer, while dissolved oxygen (DO) ranged from 6.5 to 11.5 mg L^{-1}. An inverse relationship between temperature and DO was observed, with lower DO values during summer. This pattern reflects reduced oxygen solubility at higher temperatures and increased metabolic activity of aquatic organisms.[2]

The decline in DO during warmer months corresponds with enhanced microbial decomposition of organic matter. Similar trends have been reported in Indian reservoirs under moderate anthropogenic influence, where organic inputs elevate biological oxygen demand.[3] The observed relationship indicates the combined effect of thermal conditions and organic loading on oxygen dynamics.

pH Variation

The pH ranged from 7.0 to 8.5, indicating slightly alkaline conditions typical of productive freshwater systems. Seasonal variation showed a tendency toward lower values during the monsoon.

This shift is linked to increased input of organic matter and its decomposition, resulting in the release of CO_2. Comparable seasonal pH fluctuations have been reported in tropical reservoirs influenced by runoff and catchment processes.[1,7] The stability of pH within permissible limits reflects effective buffering, while minor variations indicate external inputs associated with anthropogenic activity.

Dissolved Oxygen and Organic Load

Dissolved oxygen levels remained above 5 mg L^{-1} throughout the study period, indicating suitable conditions for aquatic life. A decline during summer suggests increased biological oxygen demand.

This reduction is associated with the accumulation of organic matter from agricultural runoff and domestic activities, which enhances microbial respiration. Similar observations have been reported in semi-urban reservoirs where diffuse anthropogenic inputs contribute to oxygen depletion without causing severe hypoxia.[8,9] The findings indicate measurable organic loading in the system.

Carbon Dioxide and Decomposition Processes

Free CO_2 concentrations ranged from 4-18 mgL^{-1}, with peak values during the monsoon season. This increase reflects intensified decomposition of organic matter and elevated respiration rates.

Monsoon-driven inflow transports organic debris and nutrients into the reservoir, influencing carbon dynamics. Similar patterns are reported in tropical aquatic systems where seasonal inflow governs CO_2 variation.[2] The elevated CO_2 levels indicate indirect anthropogenic influence through enhanced organic input.

Table 1- Annual range, mean values, and standard limits of physico-chemical parameters of Latratu Dam (2024)

Parameter	Range	Mean	Standard Range
Temperature (°C)	18–32	25	Variable
pH	7.0–8.5	7.8	6.5–9.0
DO (mg L^{-1})	6.5–11.5	8.9	>5
Free CO_2 (mg L^{-1})	4–18	10	<20
Alkalinity (mg L^{-1})	40–110	75	20–200
Chloride (mg L^{-1})	20–85	50	<250
Sulphate (mg L^{-1})	30–120	70	<200
Phosphate (mg L^{-1})	0.02–0.25	0.12	<0.5

Table 2- Seasonal variation of physico-chemical parameters in Latratu Dam (2024)

Parameter	Winter	Summer	Monsoon
Temperature (°C)	18–22	28–32	24–28
pH	7.5–8.5	7.2–8.0	7.0–7.8
DO (mg L^{-1})	9–11.5	6.5–8	7–9
CO_2 (mg L^{-1})	4–8	8–14	10–18
Alkalinity (mg L^{-1})	60–90	70–110	40–80
Turbidity	Low	Low	High
Phosphate (mg L^{-1})	0.02–0.08	0.08–0.15	0.15–0.25

Alkalinity and Buffering Capacity

Alkalinity ranged from 40 to 110 mg L^{-1}, indicating moderate buffering capacity. Seasonal variation showed a decline during monsoon due to dilution from rainfall.

Runoff from surrounding land carries dissolved ions into the reservoir, maintaining alkalinity within a stable range. Comparable values are reported in freshwater systems influenced by both natural geology and anthropogenic inputs.[1]

Chloride and Sulphate as Indicators of Anthropogenic Input

Chloride concentrations (20-85 mg L^{-1}) and sulphate concentrations (30-120 mg L^{-1}) remained within permissible limits and showed seasonal variation. Elevated values during monsoon indicate input from surface runoff.

These concentrations exceed typical baseline levels of untouched freshwater systems and reflect diffuse anthropogenic sources such as domestic waste, soil leaching, and agricultural activities. Similar enrichment patterns are observed in reservoirs with low-intensity human influence.[3,10]

Nutrient Enrichment and Eutrophication Risk

Phosphate concentrations ranged from 0.02 to 0.25 mg L^{-1}, with maximum values during monsoon. This pattern indicates nutrient loading from agricultural runoff, detergents, and organic waste.

Pristine water bodies usually exhibit phosphate levels below 0.05 mg L^{-1}. The observed concentrations suggest moderate enrichment and mesotrophic conditions. Similar nutrient trends are reported in Indian reservoirs undergoing early stages of eutrophication.[9] Continued nutrient input may accelerate eutrophication.

Interrelationship

Correlation analysis revealed a negative relationship between DO and CO_2 and positive relationships between phosphate, turbidity, and temperature. These relationships highlight the interaction among physico-chemical processes.

The inverse DO-CO_2 relationship reflects the balance between photosynthesis and respiration. The association between nutrients and turbidity indicates runoff-driven input. Similar interdependencies are widely reported in freshwater ecosystems influenced by seasonal variation and anthropogenic inputs.[2]

Anthropogenic Influence on Reservoir Ecology

The physico-chemical characteristics indicate measurable anthropogenic influence on the reservoir. Agricultural runoff contributes nutrients, domestic activities introduce organic matter, and catchment disturbances increase turbidity and chemical variability.

These effects intensify during monsoon, when surface runoff acts as the primary transport pathway. Comparable patterns are observed in semi-rural reservoirs where diffuse sources dominate water quality changes.[3] The reservoir shows early-stage ecological alteration linked to increasing human activities.

Comparative Analysis

Latratu Dam shows lower pollutant concentrations and better oxygen conditions than polluted urban lakes. Nutrient levels exceed those of pristine systems, indicating moderate anthropogenic enrichment.

The dam represents a transitional system between unpolluted and moderately impacted conditions. Such systems are common in regions experiencing gradual ecological change driven by diffuse human activities. Continuous monitoring remains necessary to prevent further degradation.

Table 3- Comparative assessment of water quality parameters indicating anthropogenic influence in Latratu dam and reference systems

Parameter	Latratu Dam	Polluted Urban Lake	Clean Reservoir
DO	High	Low	High
pH	Neutral–alkaline	Variable	Stable
Chloride	Low	High	Low
Sulphate	Moderate	Very high	Low
Nutrients	Moderate	High	Low

CONCLUSION

The study demonstrates that Latratu Dam, while currently maintaining acceptable water quality, is increasingly influenced by anthropogenic activities. Seasonal nutrient enrichment, elevated CO_2 levels, and chemical variability indicate growing pressure from agricultural runoff, domestic usage, and catchment disturbances. These changes signify the early stages of ecological stress and potential eutrophication. Without proper management, continued anthropogenic input may lead to significant degradation of water quality. Therefore, regular monitoring, catchment management, and control of nutrient inflow are essential to ensure long-term ecological sustainability.

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